Research Method-Fit Matrix (revised v1.1)

PhD in Deep Tech — UNMSM · Session 2 · Juseph Montesinos Asmat

1. Research question (refined from Session 1)

How can an *advisory* digital twin for decision support and energy-and-capacity optimization be designed for mid-sized Peruvian data centers with limited telemetry (INSUR pilot case), and how can it be evaluated — both its validity as a model and its utility for decisions — as improving operational decisions over manual management, such that its design principles are transferable to other sites?

Paradigm (Session 1): Design Science · **Epistemology:** pragmatism

2. Three candidate methods

- **A — Design Science:** build and evaluate an artifact, validated by utility.
- **B — Simulation:** model the system and run "what-if" scenarios.
- **C — Case Study (multiple):** one or more bounded sites studied in depth.

3. Fit matrix (E.D.F.C.V.)

Weights (sum to 100): E=20 · D=15 · F=20 · C=30 · V=15
Gate rule: E ≤ 2 eliminates the method (regardless of other scores).

Criterion (weight)	A · Design Science	B · Simulation	C · Case Study (multiple)	One-line justification
E · Epistemological fit (20)	5/5	5/5	3/5	DSR and Simulation are natural fits for my paradigm; multiple case requires a cited warrant (Yin multi-case, Flyvbjerg 2006).
D · Data availability (15)	4/5	4/5	4/5	Confirmed access to logs and component telemetry (XClarity, VMware, Spectrum Control, PDUs) at INSUR and a second site; no DCIM/sensors → not 5.
F · Feasibility (20)	4/5	4/5	4/5	MVP buildable and integrable; a second site adds field logistics; viable within 3 years.
C · Contribution type (30)	**5/5**	4/5	3/5	DSR yields artifact + transferable design principles (my goal). Simulation yields a novel *what-if* comparison across small Peruvian DCs. Case yields analytical

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				generalization, but describes rather than builds.
V · Venue fit (15)	**5/5**	4/5	3/5	DESRIST publishes multi-site DSR; Simulation → WinterSim; Case outside the DSR circuit.
GATE (E≤2 eliminates)	survives	survives	survives	None falls. Won on C and V, as it should.
Raw total /25	**23**	21	17	—
Weighted total /500	**465**	420	335	The high weight on C widens DSR's lead (a sign of honest weighting).

4. Defending the weights

Highest weight — C (30): "I weighted contribution highest because a doctorate is justified by producing new, verifiable knowledge — a resolved *gap* — not by building an application; without an original contribution there is no doctoral thesis."

Lowest weights — D (15) and V (15): "I weighted data availability low because I already have confirmed access to INSUR and a second site; data is not my bottleneck. And I weighted venue low because the contribution defines where to publish, not the reverse."

On E (20), when probed: "I did not give E the top weight because E is already protected by the *gate* (E≤2 eliminates): I gave it the highest power in the matrix, the power to discard. And I weight F (20) because real feasibility within three years is a hard filter that removes unbuildable projects."

5. Winning method: Design Science

I choose Design Science because my contribution is to **build and validate an artifact with demonstrated utility** (an *advisory* digital twin), and because only DSR produces the kind of knowledge my question demands: an artifact plus transferable design principles, evaluated in real operations and generalizable across sites. I reject **Simulation** because it predicts scenarios without demonstrating utility in real operations; I reject **Case Study** because it describes the site without building a solution.

*(Gregor & Hevner, 2013: the contribution type is **exaptation —a known solution applied to a new domain—, not invention.)***

6. Exit commitment

> "I will use **Design Science** because my contribution is an artifact validated by utility, and I reject **Simulation** because it does not demonstrate utility in real operations, and **Case Study** because it does not build a solution."

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